

Cooling *as a Service*

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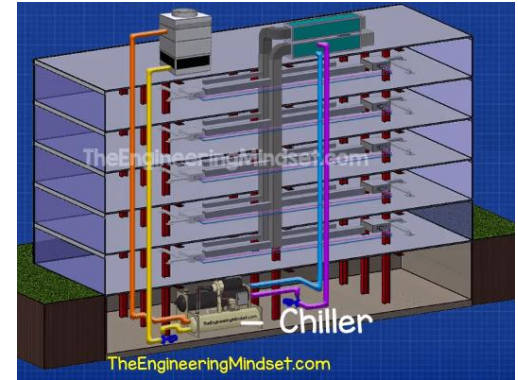
How CaaS Works



Thermal Energy Storage

Chill water at off-peak hour when electricity tariff is low. Discharge chilled water to the clients building during peak hour.

Mostly underground pipes



Client Buildings

Heat exchanger will then cool the internal building water which is then pumped to Air Handling Units on various floors.



On-Site CaaS

Decentralized

- Equipment inside your building
- Serves strictly to a specific building/facility
- Localized chillers and pumps
- Moderate lock-in level



District Cooling

Centralized DCS

- Central chiller plant, off-site
- Serves an entire precinct/campus
- From a centralized tank to buildings via underground pipes
- Higher lock-in level due to heavy civil work

Pricing Mechanism

RM per Refrigerant Ton-Hour

Variable Fees:

$RT = \text{Cooling Load (kW)} / 3.517$

$RTh = RT \times \text{Hours}$

*RM 0.45-0.60 per RTh

1 RT = 3.517 kW

Fixed Fees:

Maximum cooling capacity contractually reserved per month

Monthly Bill:

Fixed + Variable Fees

Capacity Reservation

Fixed Floor:

Fixed Fee = Cooling Capacity Reserved, $P1 \times \text{Capacity Rate, } R1$

Variable Top-Up:

Variable Fee = Consumption beyond Reserve, $P2 \times \text{Variable Rate, } R2$

Monthly Bill:

Fixed + Variable Fees

RM per kW of IT Load Cooled

Monthly Bill:

Monthly Fee = Contracted Rate (RM/kW) $\times$ Server Load (kW)

Why CaaS Exists



01

Zero CAPEX

Cooling infrastructure doesn't drive core revenue — CaaS eliminates this balance sheet burden.



02

Precision Cooling

Data centers demand cooling beyond standard A/C. CaaS providers fill the technical gap.



03

ESG Compliance

Running in-house cooling is costly in capital and engineering headcount. CaaS transfers this.

Market Players

KJTS Group

KIP Mall · RM 23M · 20yr

Kaer (Singapore)

TRX · RM 150M · 15yr

Engie (France)

Sunway · 24yr

Gas District Cooling

KLCC · Putrajaya

Acquire (RM 65.5 million)

Malakoff Utilities

KL Sentral

Megajana

Cyberjaya DCS

Jana DCS

Kota Iskandar, Johor

TNB Engineering Corporation

KLIA

Case Study Matrix

Project / Location	Operator	Architecture	Est. Value	Duration
Exchange TRX	Kaer	On-Site CaaS	-RM 150M	15 yrs
KIPMall (8 Malls)	Lestari / KJTS	On-Site CaaS	-RM 23M	20 yrs
KLIA	CESSB (TNEC+MAHB)	District Cooling	-RM 183M	20 yrs
Sunway	Engie-Sunway JV	DCS + TES	Undisclosed	24 yrs
KL Sentral	Malakoff Utilities (Acquired by KJTS)	District Cooling	RM 65.5M	Long-term